Linear Discriminant Functions

Why?

- computing probability density is difficult and often incorrect
- we try to calculate parameters of **discriminant function**
- much simpler classifier at the cost of insignificant decrease of classification quality
- simple formula:

$$g(x) = w^T x + w_0$$

  - w - weight vector (that we try to calculate)
  - x - input vector
  - w_0 - bias vector
- augmented feature vector:

$$a^T = [w_0 w_1 ... w_d] = [w_0 w^T]$$

- discriminant function:

$$g(x) = \sum_{i=0}^{d} w_i w_x = a^T y$$

- in general case we have c discriminant functions - one for each class
- memory needs for one classifier:
  - D - number of features
  - C - number of classes
  - for one classifier we need $D + 1$ memory (all features + bias)
  - all memory needs depends on solution type (one v all, two v all etc.)

Linearly Seperable Case

- if exists such vector a that all training samples can be classified correctly, the samples are *linearly seperable*
- in that case for every y_i belonging to class c_1 we have $a^T y_i > 0$
  - and for every y_j belonging to c_2 we have $a^T y_j < 0$
- we try to maximize distance between samples and seperating plane, to hope that vector will classify new samples better

Gradient methods

- we take criterion function $J(a)$
- we start at selected a_{start}
- in each step we calculate gratient vector $\nabla J(a)$ and move the solution in the directon of the steepest descent:

$$a(k+1) = a(k) - \eta(k) \nabla J(a(k))$$

  - $\eta(k)$ - learing rate
  - we usually try to decraese value of learing rate (for example proportionally to $\frac{1}{k}$)

More than two classes

One vs many/rest

- if we have $c > 2$ classes, we can prepare c linear classifiers
- **one vs. rest** approach

- rarely one class can be separated by hyperplane
- we have very unbalanced sizes of positive and negative samples

One vs one

- we prepare classifiers for every pair of classes(
- better predictions, but many more classifiers $\frac{c(c-1)}{2}$)
- result is read by looking which class got the most votes
- the problem is that each classifier is trained on parts of training set (2 classes of c), but it votes on every classification

Support vectors

- extension of linear discriminant functions
- we demand such a separating plane that all training vectors are not closer to plane than some distance $\frac{d}{2}$
- such a separating plane is determined by few vectors - **support vectors**

Linear Classifier size